A Tutorial on Graph RAG & Meta KDD Cup

2026.5.22

Data Systems Lab

Overview

- Introduction
  - Retrieval-augmented Generation (RAG)
  - Graph RAG
- Meta KDD Cup
  - Benchmark
  - Baseline Approach

DATABASE LAB @POSTECH

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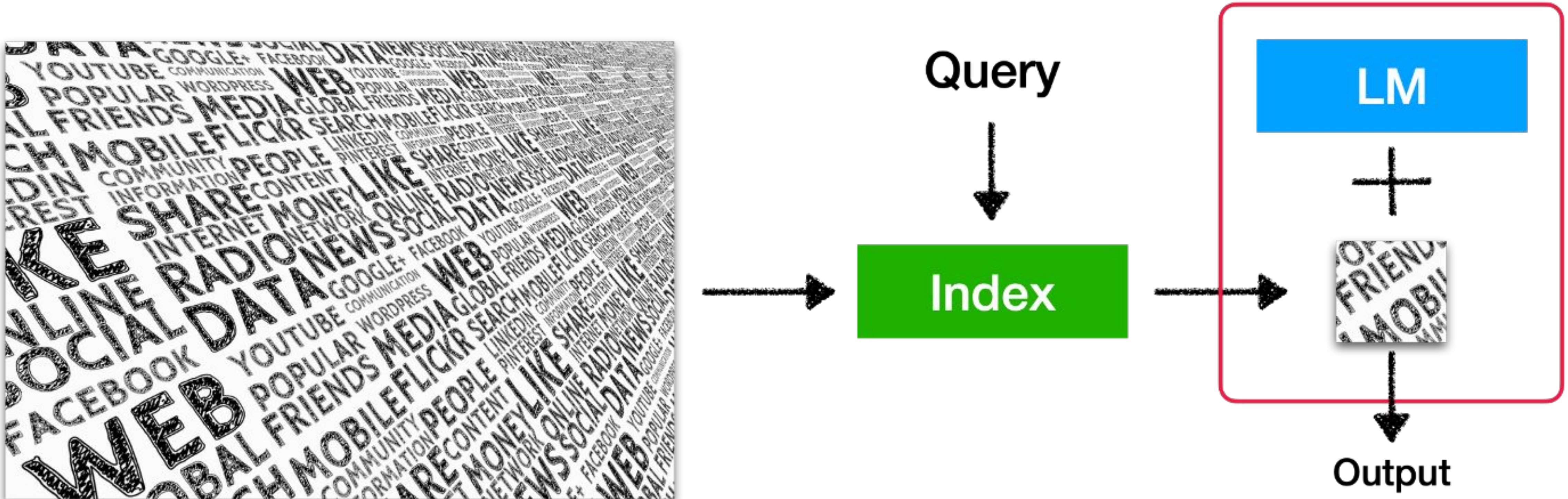
Retrieval-augmented Generation (RAG)



Datastore

Incorporate Datastore at inference

Slides adapted from the ACL-23 tutorial:
Retrieval-based Language Models and Applications

RAG Mitigates Weakness of Large Language Models

Hallucinations

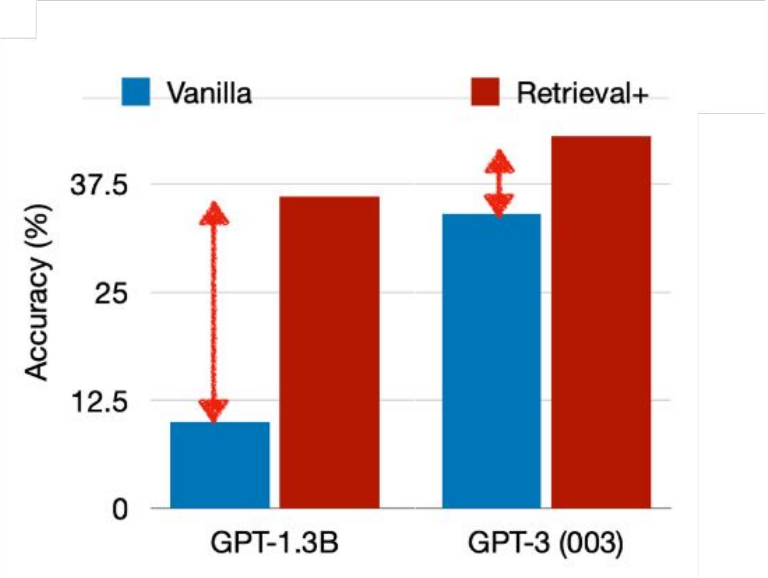
Costs of adaptations

Copyright / privacy

Large parameter size

Significant improvements across model scale, with larger gain with smaller LM

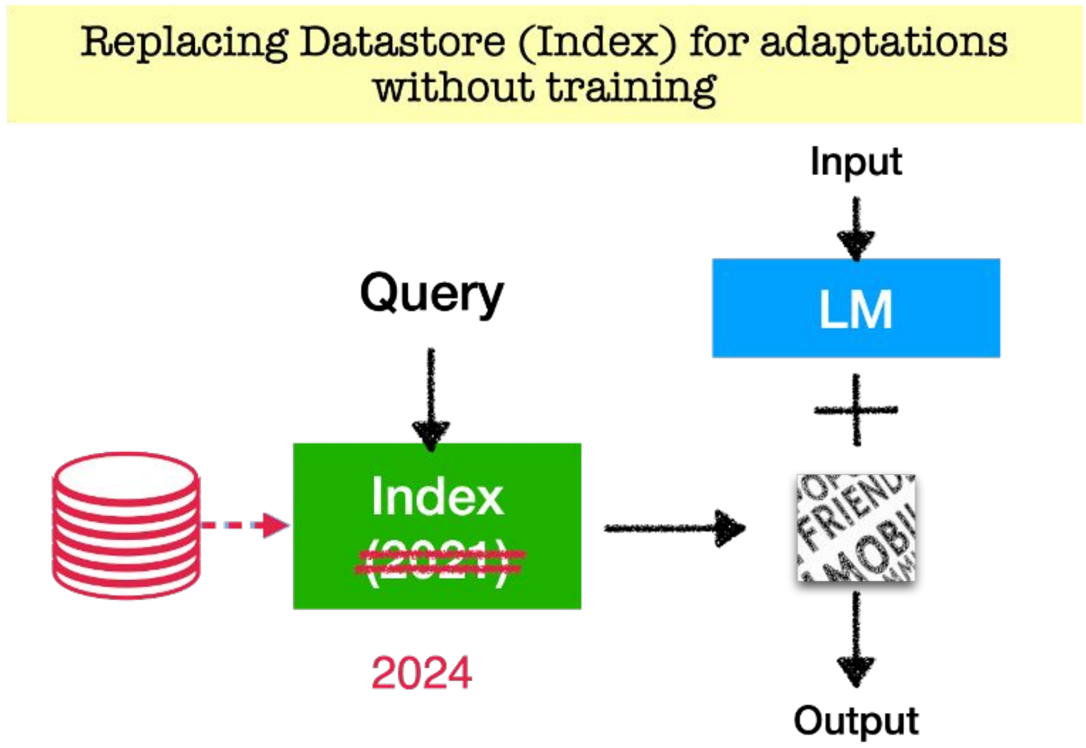
QA



Mallen*, Asai* et al., When Not to Trust Language Models: Investigating Effectiveness of Parametric and Non-Parametric Memories (ACL, Best Video; Oral) 2023.

RAG Mitigates Weakness of Large Language Models

- Hallucinations
- **Costs of adaptations**
- Copyright / privacy
- Large parameter size



Izacard*, Lewis* et al., Atlas: Few-shot Learning with Retrieval Augmented Language Models. JMLR 2023.

RAG Mitigates Weakness of Large Language Models

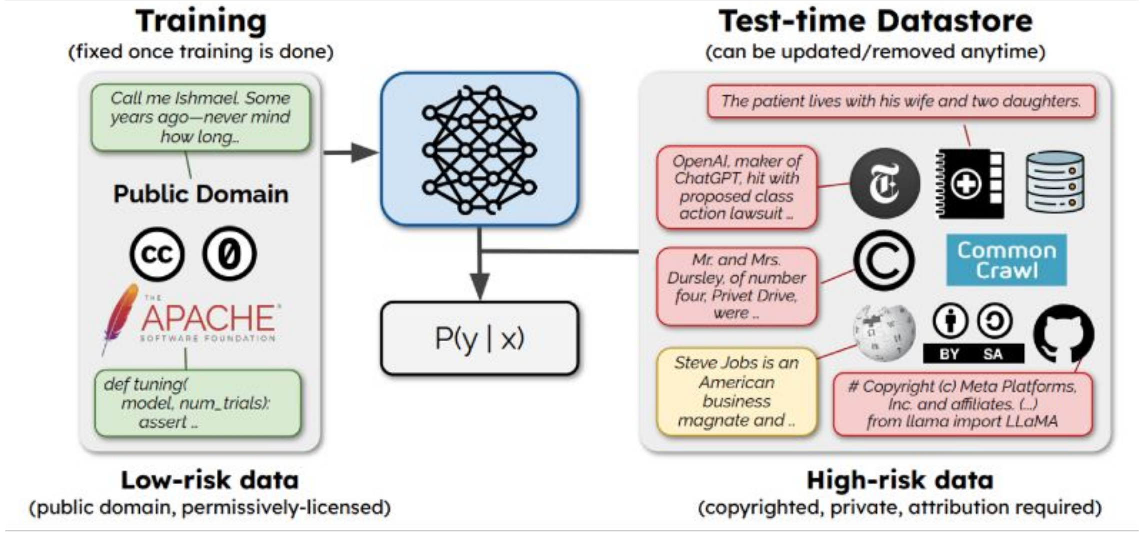
Hallucinations

Costs of adaptations

Copyright / privacy

Large parameter size

Segregating copyright-sensitive data from pre-training data



Min* and Gururangan* et al., SILO Language Models: Isolating Legal Risk In a Nonparametric Datastore. ICLR 2024

RAG Mitigates Weakness of Large Language Models

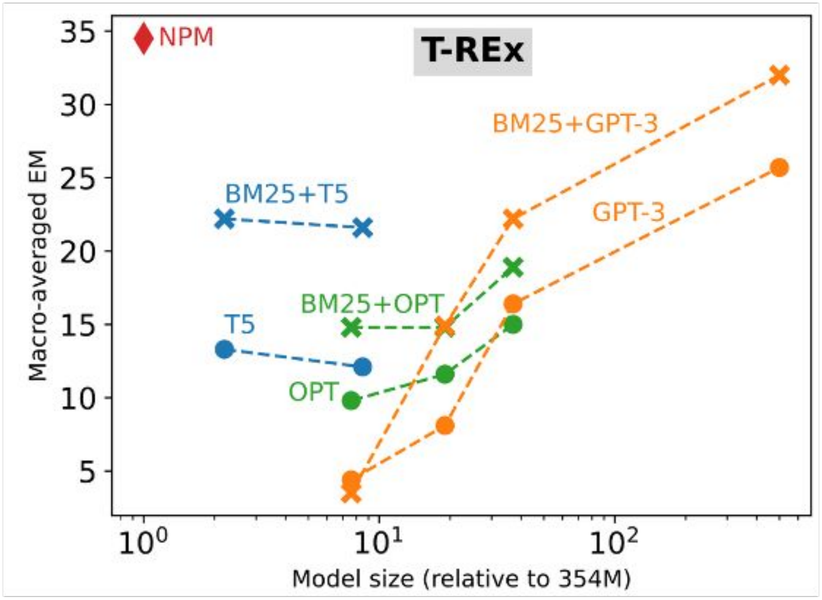
Hallucinations

Costs of adaptations

Copyright / privacy

Large parameter size

Models with much less parameters can outperforms much larger models!



Min et al., Nonparametric Masked Language Modeling. Findings of ACL 2023.

One of the Hottest Topics in VLDB 2024: (Graph)RAG

Vector Databases

A2 Panel Vector Databases: What's Really New and What's Next?

A1 Data management and support for ML/AI

Experimental Analysis of Large-scale Learnable Vector Storage Compression
University); Penghao Zhao (Peking University); Xupeng Miao (Carnegie Mellon Uni (Peking University); Bin Cui (Peking University)*

SingleStore-V: An Integrated Vector Database System in SingleStore
Zhang (Purdue University - West Lafayette); Sasha Podolsky (SingleStore); Zhou Sun (SingleStore); Robert Walzer (SingleStore); Jianguo Wang (Purdue

Chat2Data: An Interactive Data Analysis System with RAG, Vector Databases and LLMs *xi*
*Guoliang Li (Tsinghua University)**

Graph Databases

NebulaGraph

Industry Talk: Integrating GenAI with Graph: Innovations and Insights from NebulaGraph

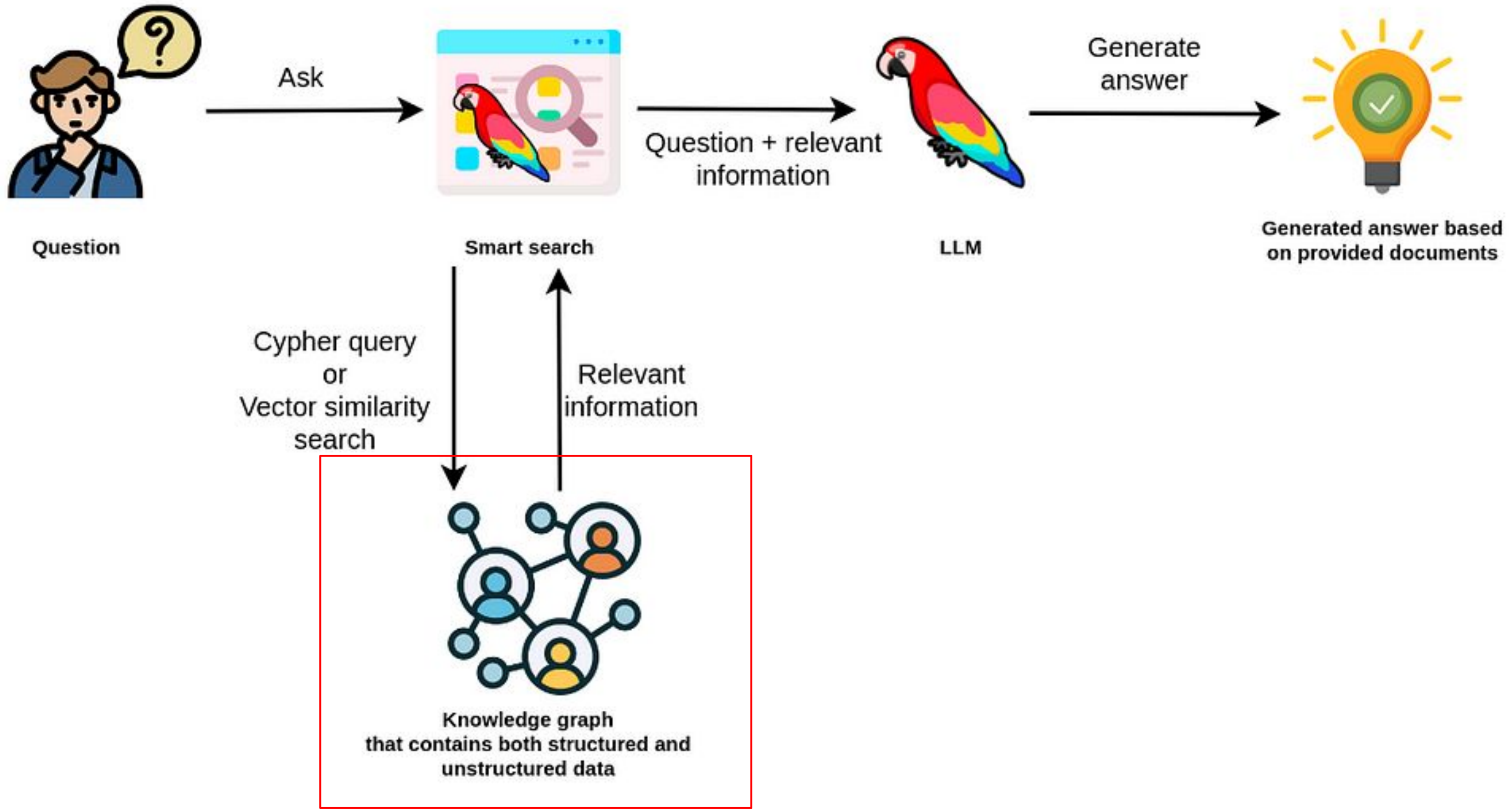
Siwei Gu & Yihang Yu (NebulaGraph, China)

LLM+KG Workshop

neo4j

Microsoft

GraphRAG Overview



GraphRAG Overview (From Neo4j Document)

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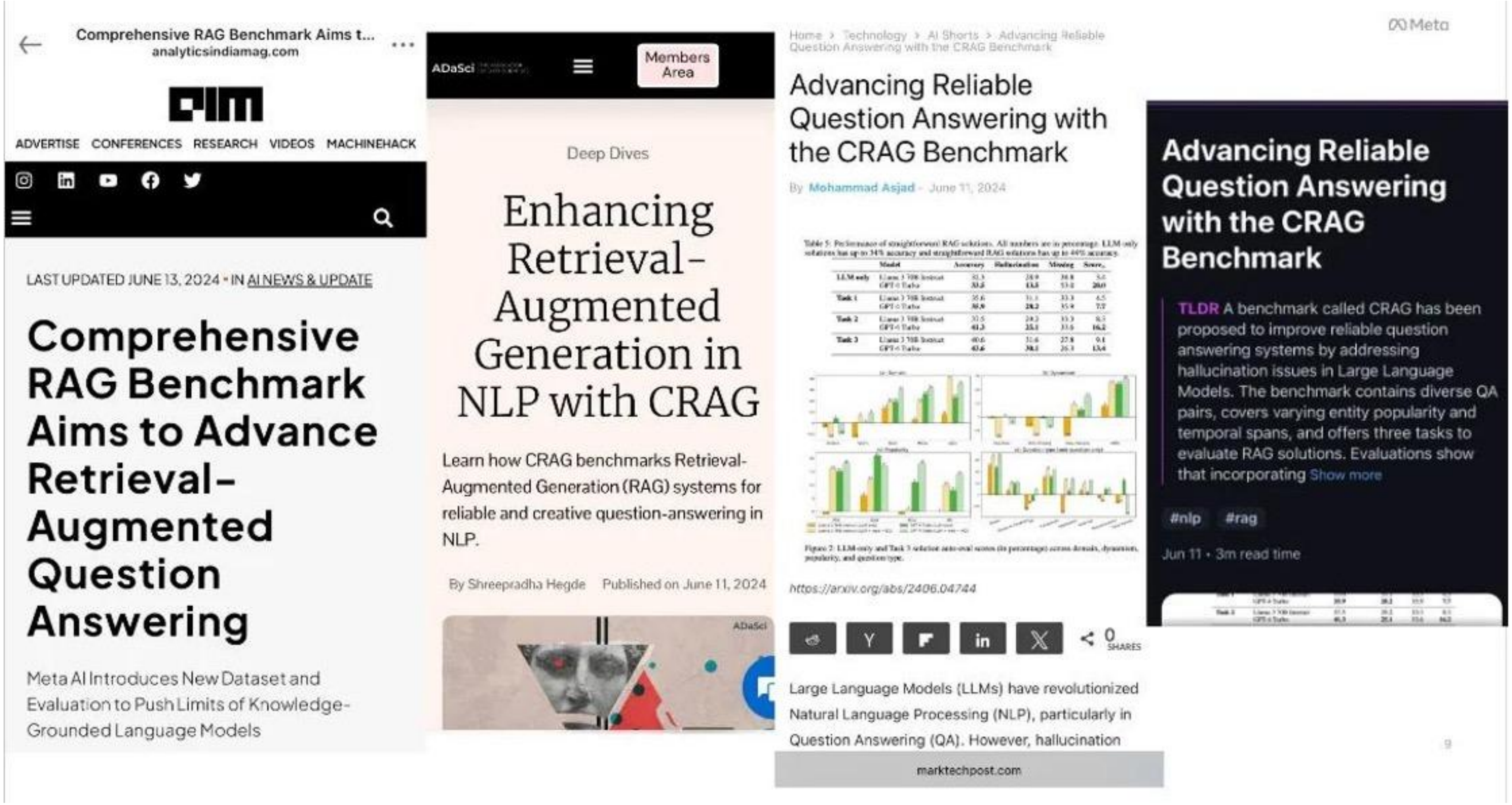
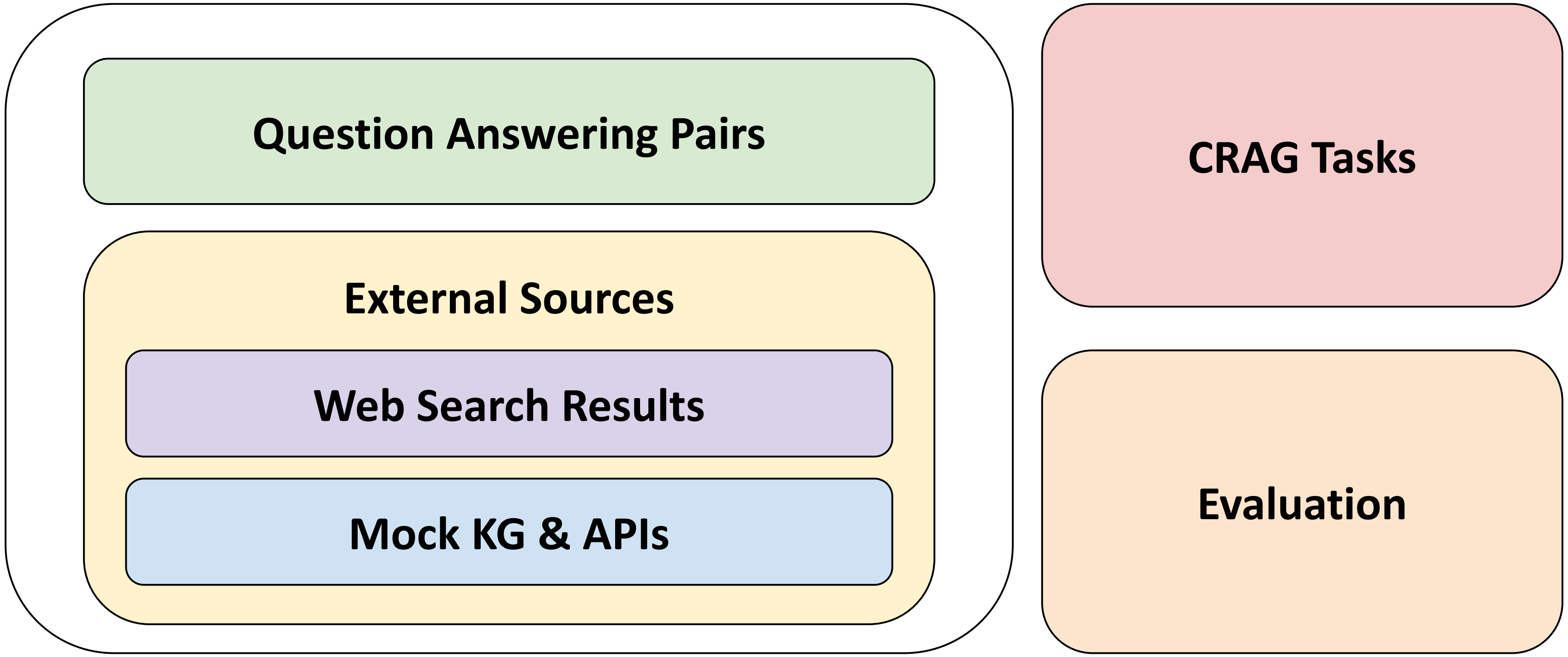
Meta KDD Cup 2024

- 2300+ participants, 384 teams, 5600+ submissions
- CRAG benchmark was listed in HuggingFace "Daily papers".
- Our team was the only Korean team to receive an award, achieving First Place in the Comparison Question category for Tasks 1, 2, and 3!!



Comprehensive RAG Benchmark Aims t...
analyticsindiamag.com
ADVERTISE CONFERENCES RESEARCH VIDEOS MACHINEHACK
LAST UPDATED JUNE 13, 2024 • IN AI NEWS & UPDATE
Comprehensive RAG Benchmark Aims to Advance Retrieval-Augmented Question Answering
Meta AI Introduces New Dataset and Evaluation to Push Limits of Knowledge-Grounded Language Models
ADaSci
Members Area
Deep Dives
Enhancing Retrieval-Augmented Generation in NLP with CRAG
Learn how CRAG benchmarks Retrieval-Augmented Generation (RAG) systems for reliable and creative question-answering in NLP.
By Shreepradha Hegde Published on June 11, 2024
Home > Technology > AI Shorts > Advancing Reliable Question Answering with the CRAG Benchmark
Advancing Reliable Question Answering with the CRAG Benchmark
By Mohammad Asjad - June 11, 2024
https://arxiv.org/abs/2406.04744
0 SHARES
Large Language Models (LLMs) have revolutionized Natural Language Processing (NLP), particularly in Question Answering (QA). However, hallucination
marktechpost.com
Meta
Advancing Reliable Question Answering with the CRAG Benchmark
TLDR A benchmark called CRAG has been proposed to improve reliable question answering systems by addressing hallucination issues in Large Language Models. The benchmark contains diverse QA pairs, covers varying entity popularity and temporal spans, and offers three tasks to evaluate RAG solutions. Evaluations show that incorporating Show more
#nlp #rag
Jun 11 • 3m read time

CRAG Benchmark Overview



CRAG Benchmark Overview

Question Answering Pairs

External Sources

Web Search Results

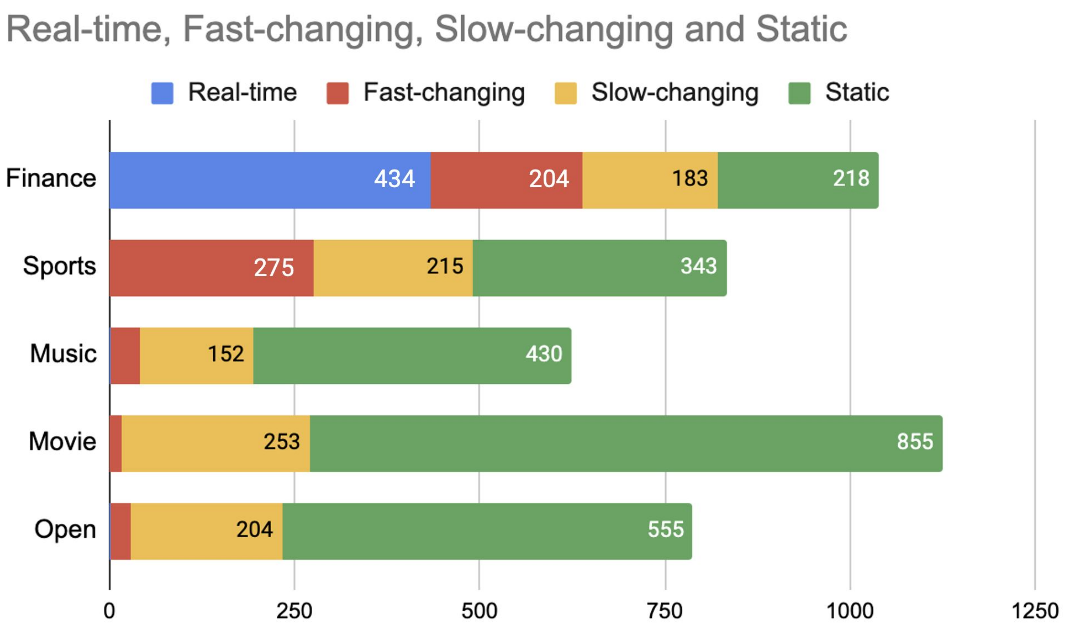
Mock KG & APIs

CRAG Tasks

Evaluation

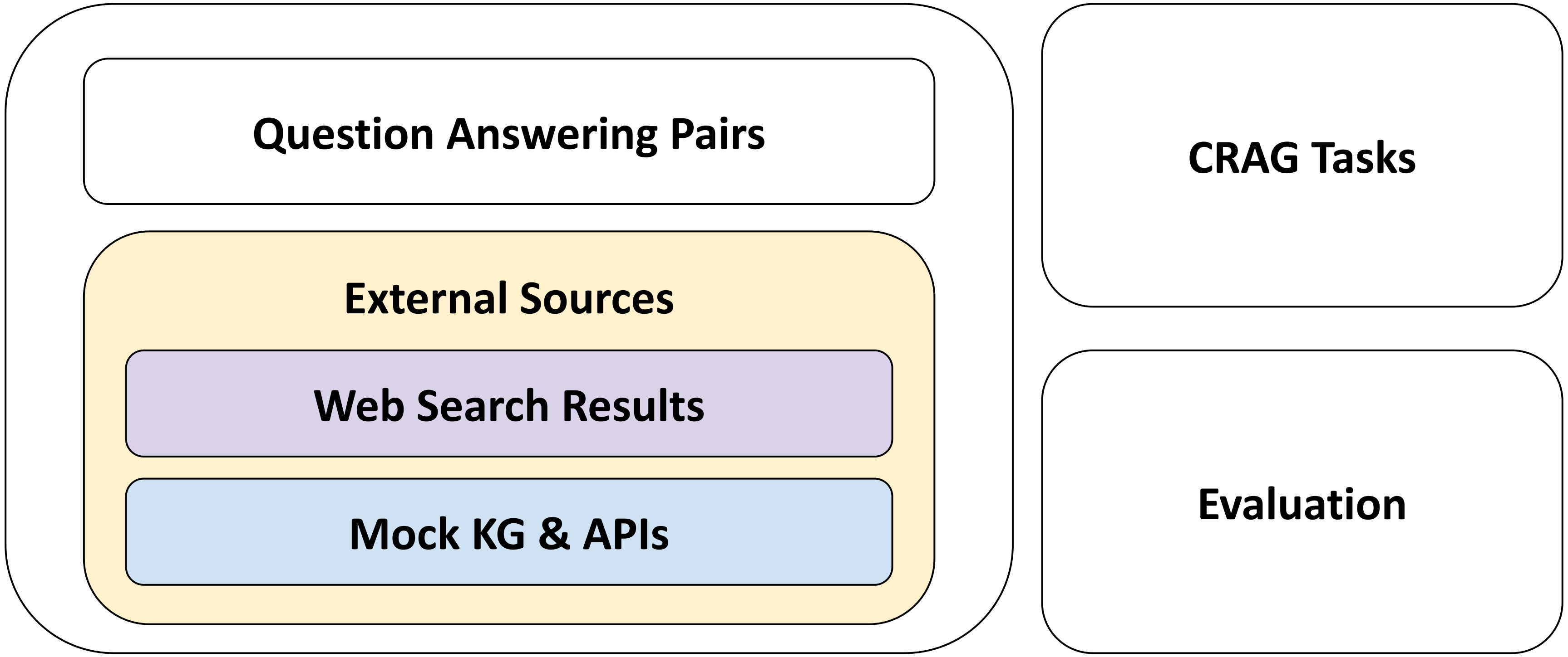
Question Answering Pairs

- 4400+ QA pairs from 5 domains (Finance, Sports, Music, Movie, Encyclopedia)
- Questions for static, slow-changing, fast-changing, and real-time information
- Questions for head, torso, and tail entities
- Simple-fact questions and complex questions



Total	Simple	Simple w. Cond	Set	Comparison	Aggregation	Multi-hop	Post-Processing	False Premise
4409	1205	689	403	546	489	382	180	525

CRAG Benchmark Overview



External Sources

- Web Search Results: 50 webpages for each question from BraveAPI web search
- Mock KG & APIs
  - Mock KG: 2.6M entities
  - Mock APIs: 38 mock APIs

Q1:
What's the latest film that walt becker has directed?
Q2:
Which one of these came out earlier, the greater meaning of water or small town ecstasy?

API for Q1:
```
get_movie_person_crew(None,"walt becker", eq(job, "Director"));
sort(None,-year)["movie_name"]
```
API for Q2:
```
get_movie("greater meaning of water")["release_date"];
get_movie("small town ecstasy")["release_date"]
```

CRAG Benchmark Overview

Question Answering Pairs

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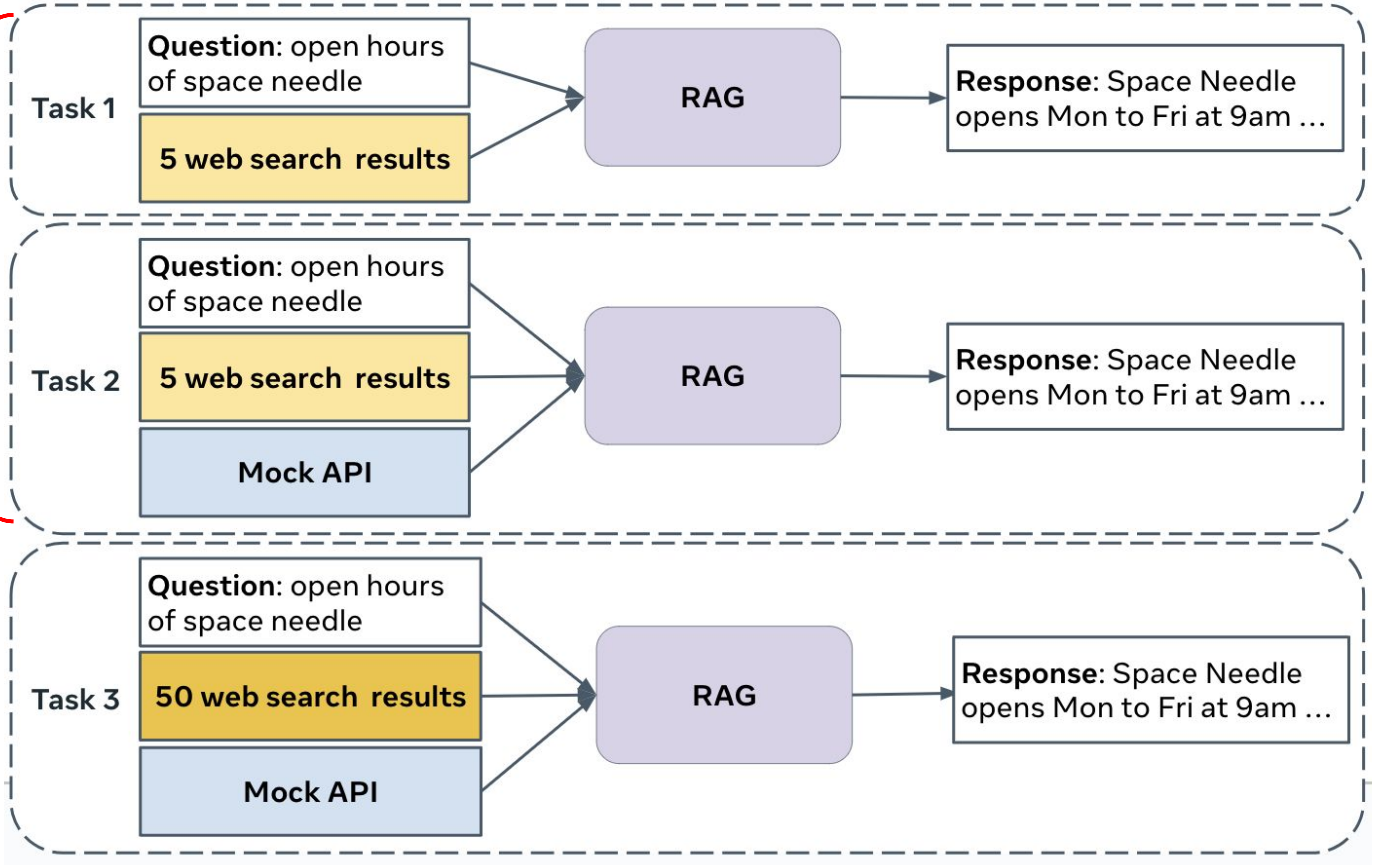
CRAG Tasks

Evaluation

CRAG Tasks

Three tasks build up information gradually to test different capabilities of RAG systems.

Today's focus



CRAG Benchmark Overview

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CRAG Tasks

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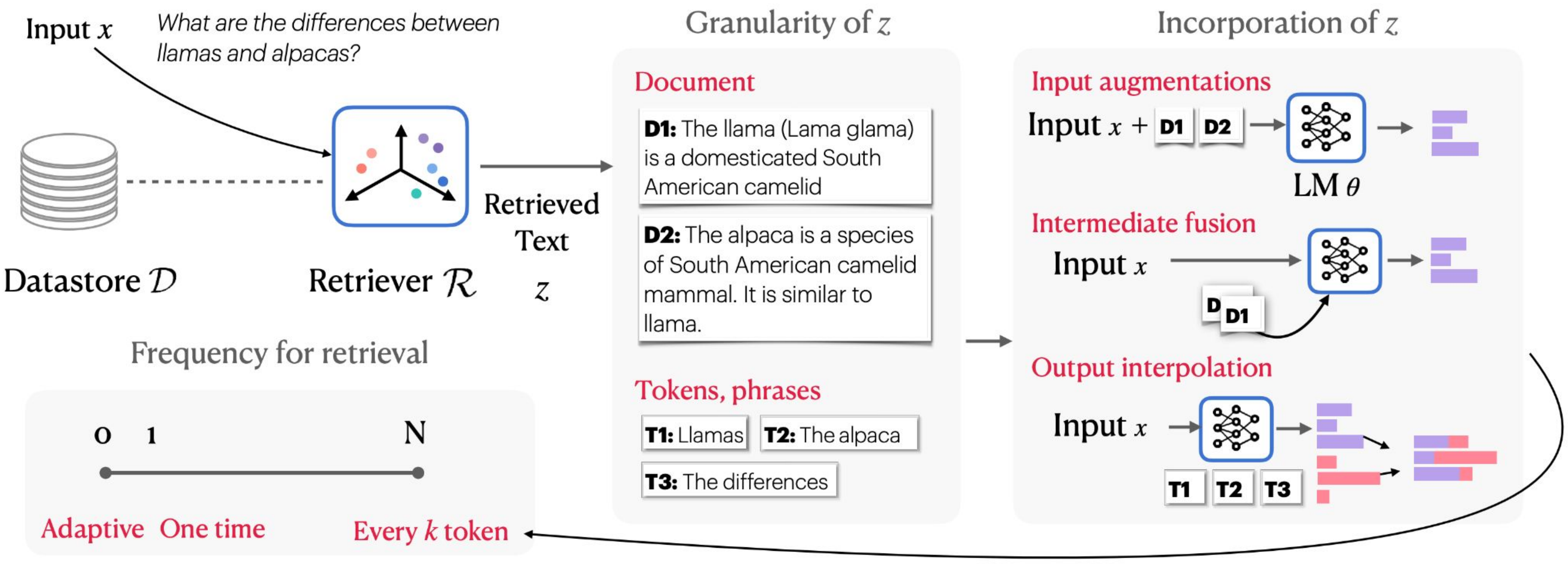
Evaluation

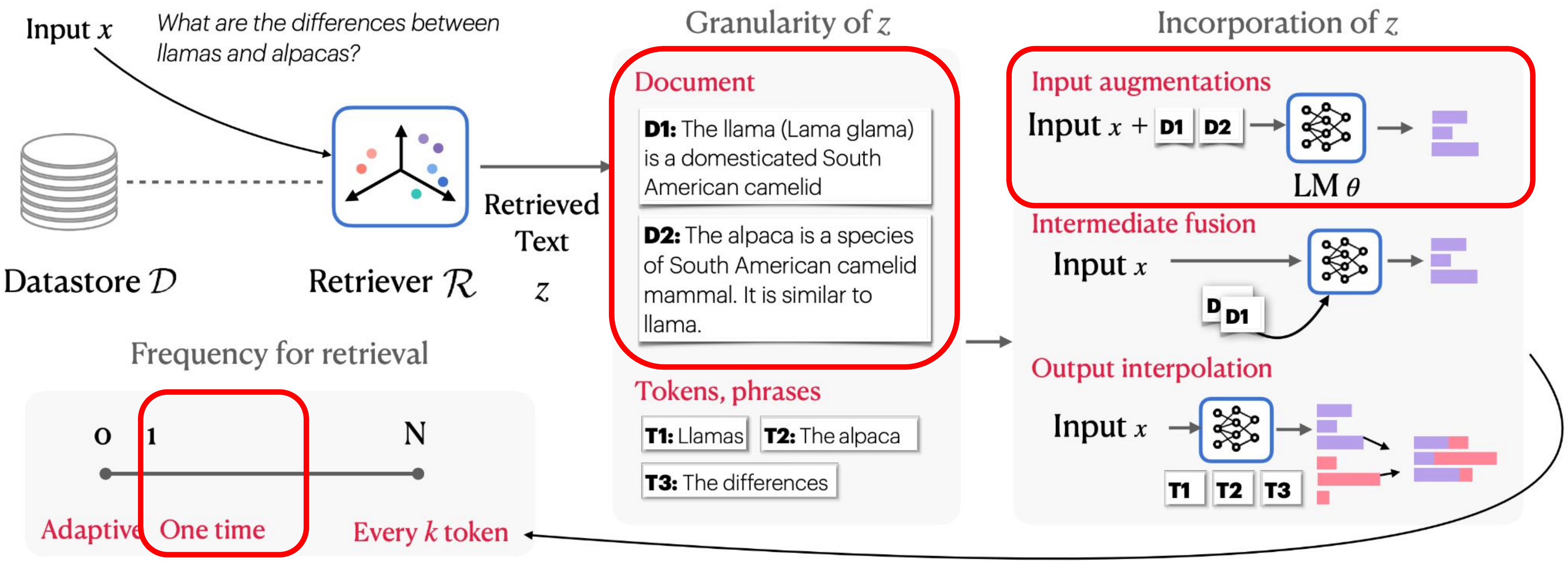
- Metrics: CRAG Score = Exact Accuracy + 0.5 * Accuracy - Hallucination rate
  - **Exact Accuracy:** The percentage of questions for which the generated answer exactly matches the ground truth answer.
  - **Accuracy:** The percentage of questions for which the generated answer is not exact but has the same meaning as the ground truth.
  - **Hallucination:** The percentage of questions for which an incorrect answer was generated.
  - **Missing:** The percentage of questions where the response was "I don’t know."
- Evaluation
  - Auto-eval (with GPT-4)
  - Manual-eval (with Human)

Diverse Architectures of RAG



Asai, Akari, et al. "Reliable, adaptable, and attributable language models with retrieval." arXiv preprint arXiv:2403.03187 (2024).

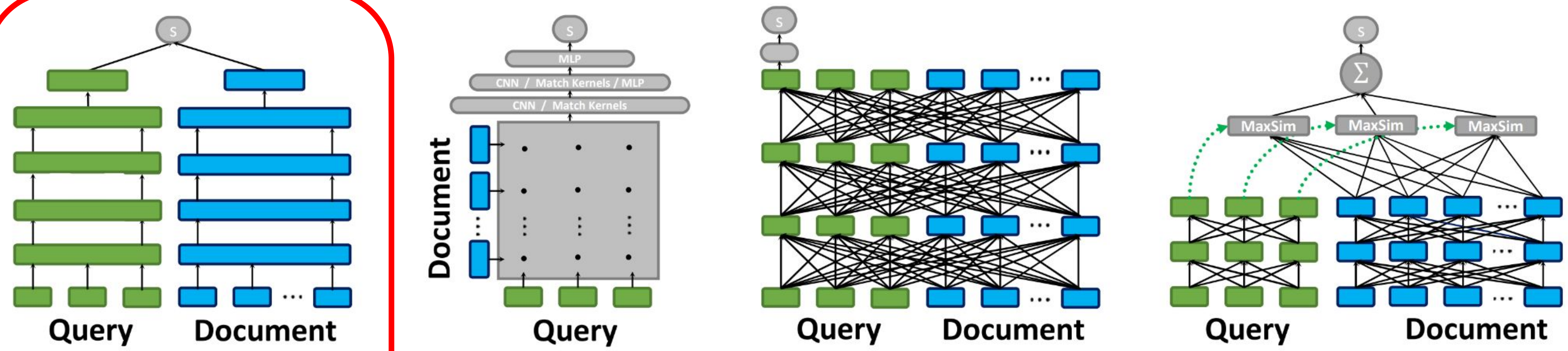
Baseline Approach



Asai, Akari, et al. "Reliable, adaptable, and attributable language models with retrieval." arXiv preprint arXiv:2403.03187 (2024).

Retrieval Module for Web Search Results

- Retriever performs representation-based similarity search.
- It retrieves top-k relevant sentences.



(a) Representation-based Similarity
(e.g., DSSM, SNRM)

(b) Query-Document Interaction
(e.g., DRMM, KNRM, Conv-KNRM)

(c) All-to-all Interaction
(e.g., BERT)

(d) Late Interaction
(i.e., the proposed ColBERT)

Mock APIs & Mock KG

- Task 2 provides mock APIs to query the provided mock knowledge graph (mock KG).

```
Q1:
What's the latest
film that walt becker
has directed?
Q2:
Which one of these
came out earlier, the
greater meaning of
water or small town
ecstasy?
```

```
API for Q1:
get_movie_person_crew(None,"wal
t becker", eq(job, "Director"));
sort(None,-year)["movie_name"]
API for Q2:
get_movie("greater meaning of
water")["release_date"];
get_movie("small town
ecstasy")["release_date"]
```

- The mock KG, as a structured knowledge base, offers precise information; however, generating an accurate query is essential for retrieving correct answers.

Retrieval Module for Mock KG

- The knowledge graph retrieval module follows these three steps:
  a. The LLM generates the query domain and API arguments.
  b. Based on the generated query domain and API arguments, a decision tree is used to sequentially call the appropriate mock APIs.
  c. The results from these mock API calls are provided to the LLM along with retrieved results from web search results.
- We aim to go a step further by leveraging the Model Context Protocol (MCP) to upgrade the knowledge graph retrieval module.

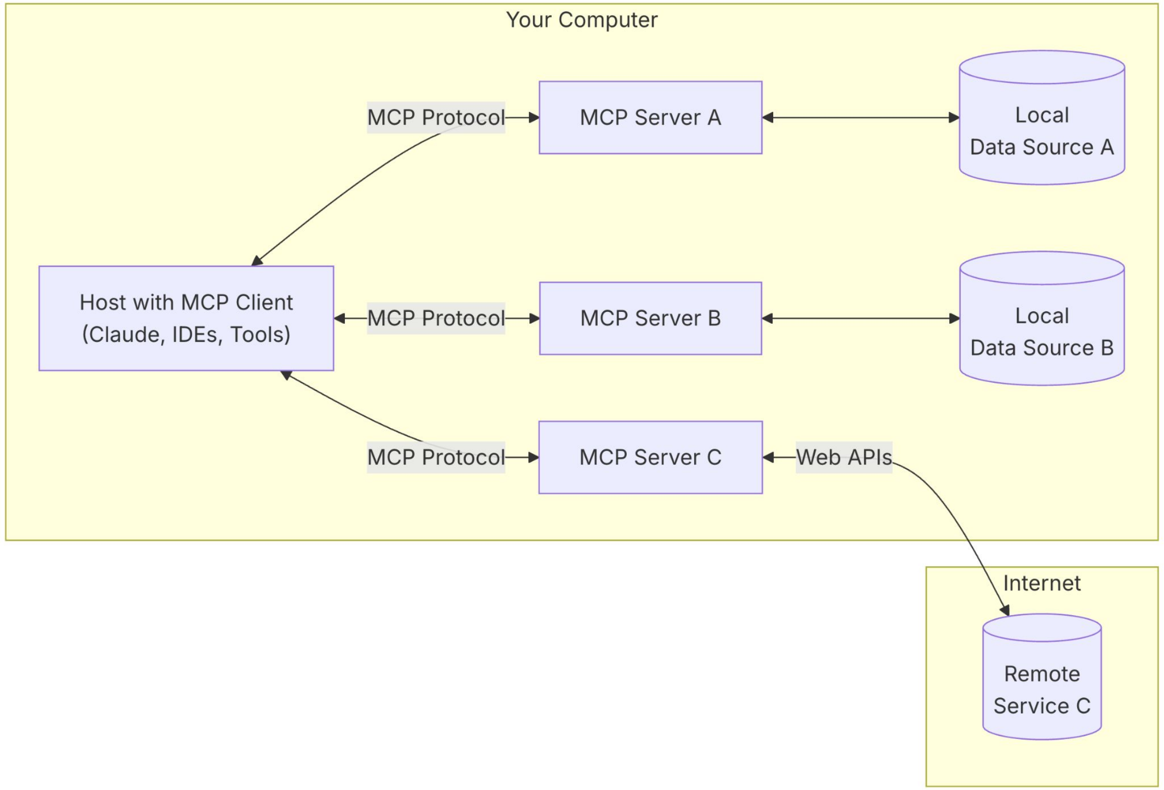
Model Context Protocol

Model Context Protocol (MCP) is an open protocol that standardizes how applications provide context to LLMs.

- Think of MCP like a USB-C port for AI applications.
- Just as USB-C provides a standardized way to connect your devices to various peripherals and accessories, MCP provides a standardized way to connect AI models to different data sources and tools.

General Architecture

At its core, MCP follows a client-server architecture where a host application can connect to multiple servers:



Appendix